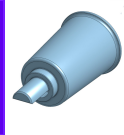
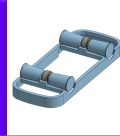




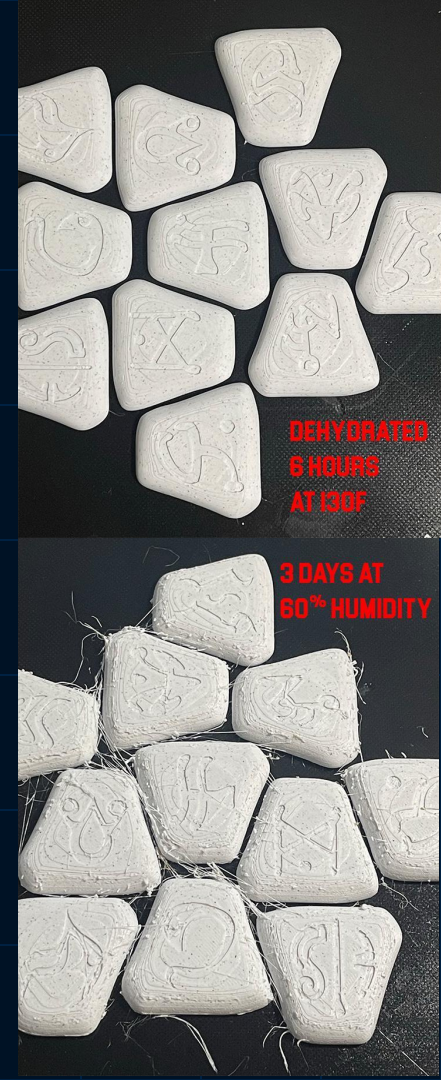
Term Project: Filament Dry Box



Josh Kieler
Fall 2025

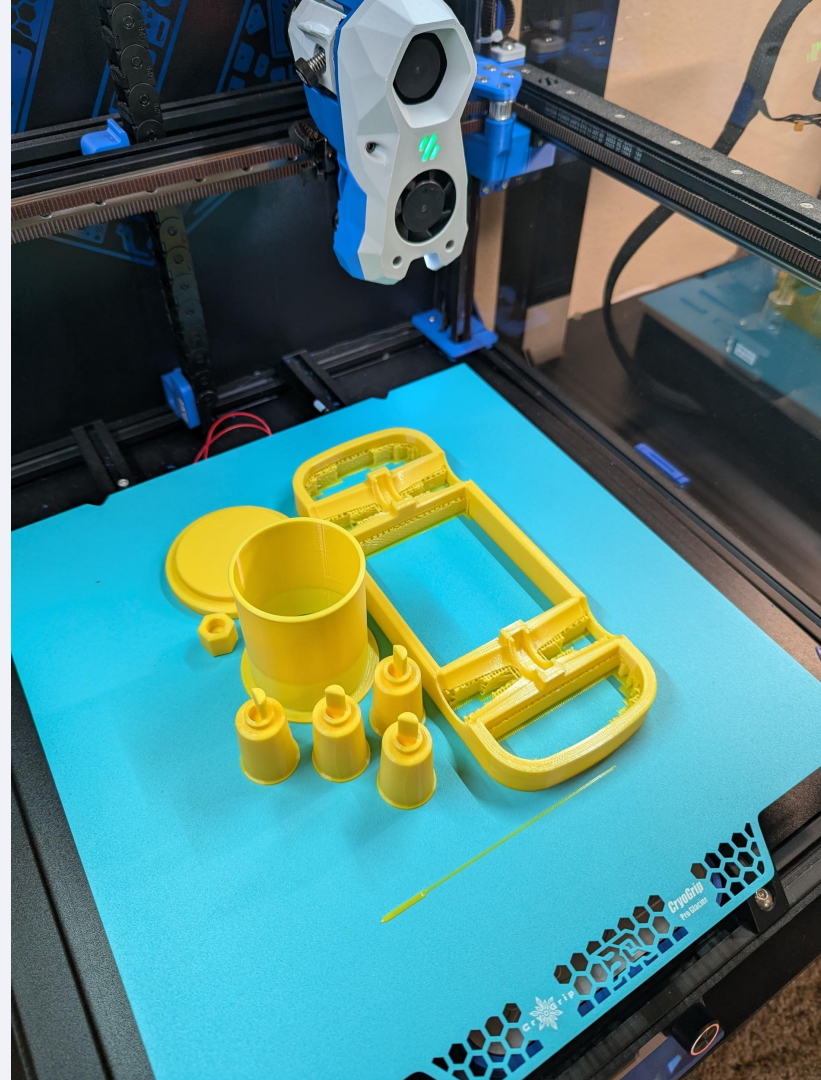
The Problem

- Filament absorbs moisture from the air.
- Wet filament does not print well
 - Poor surface quality
 - Loss of dimensional accuracy
 - Stringing
- Filament may be dried
- Filament needs to be maintained in a dry environment while printing
- Various solutions exist, but could fit my setup better



Design Goals

- Reduce required hardware
- Reduce part number for easier assembly
- Incorporate a reusable desiccant container
- Use a readily available inexpensive snaplock container as the base



Design challenges



Reverse engineering
the lid profile

The injection molded lid
has a compound curve at
either end.

Fitting the skate
bearing

A 608 skate bearing
needed to be incorporated
into the base for smooth
motion of the roller axles.





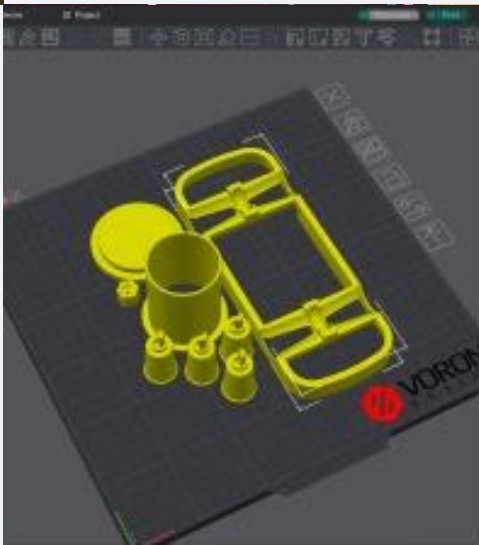
Base Support

Printing the base in one piece requires significant support. Settings needed to be optimized.

Printing Considerations

Bridge Settings

The lid seal has a large bridged area. The bridge angle may need to be adjusted.



Thank You!

Files available upon request.
Email: Josh.Kieler@PCC.edu

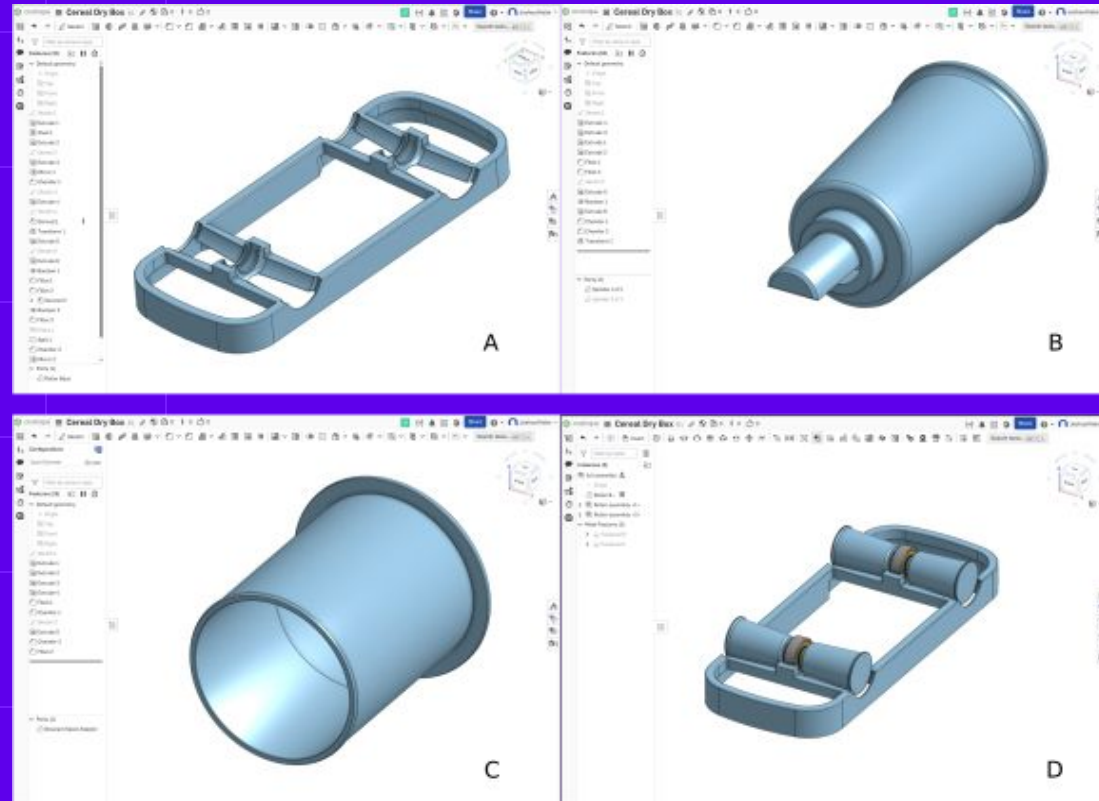


Figure 2: Screenshots of CAD Models from OnShape: A. base, B. roller half, C. dessicant spool adapter, D. Assembly